

March 16, 2022

Virtual Session.



Comments on the proof of Chebyshev:

$$(-\infty, \infty) = \{x \in \mathbb{R} : g(x) \geq r\} \cup \{x \in \mathbb{R} : g(x) < r\}$$

Two mutually exclusive sets

$$\int_{-\infty}^{\infty} g(x) f_x(x) dx = \int_{\{x: g(x) \geq r\}} g(x) f_x(x) dx + \int_{\{x \in \mathbb{R} : g(x) < r\}} g(x) f_x(x) dx$$

$$\geq \int_{\{x: g(x) \geq r\}} g(x) f_x(x) dx$$

$$\geq \int_{\{x: g(x) \geq r\}} r f_x(x) dx$$

} because $g(x) \geq r$ in
the appropriate domain

$$= r \int_{\{x: g(x) \geq r\}} f_x(x) dx = r P(g(X) \geq r)$$

Convergence in probability

$|X_n - c|$ = "distance between X_n and c "

$$\Pr(\text{Distance between } X_n \text{ \& } c \text{ is larger than } \varepsilon) = P(n, \varepsilon)$$

$\uparrow$
real number

$$\lim_{n \rightarrow \infty} P(n, \varepsilon) = 0 \quad \text{For any } \varepsilon$$

Comments on Corollary 4.2

$E[T_n]$ is a real number

$$E[T_n] = \mu_n$$



$$\lim_{n \rightarrow \infty} \mu_n = c$$

$$\text{Var}(T_n) = \sigma_n^2$$

$$\lim_{n \rightarrow \infty} \sigma_n^2 = 0$$

$$\text{Var}(T_1) > \text{Var}(T_2) > \text{Var}(T_3) > \dots$$

